

Auditable Instruments Dominate Language Models on Calibrated Geopolitical Assessment

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Abstract

If an LLM can draft a geopolitical brief, does it *replace* the scoring stack behind serious OSINT? We show that it does not. On three frozen event packs, we pit **Claude Haiku 4.5** and **GPT-4o-mini** (temperature 0, schema-locked JSON) against ARGUS’s open instruments—absolute country risk, rate anomalies, offline ACH, and evidence-balance confidence caps—using identical inputs. **Where assessment requires calibration, instruments dominate.** Absolute risk-band agreement collapses to 0.11 for both models. ACH cell agreement stays near chance-to-middling (**Claude** 0.40, **GPT** 0.53). Confidence caps are violated by GPT (0.33 **overconfident**); Claude is better restrained but still imperfect. Models only look strong on **easy** axes: anomaly labels (1.00) and risk *rank* (GPT $\rho = 1$). Ordinal fluency is not calibrated assessment. We conclude that auditable instruments should **own** bands, ACH cells, and confidence; LLMs may assist triage and prose. Full replication: research/llm-eval/.

Keywords: OSINT evaluation, geopolitical risk, ACH, LLM evaluation, calibration, auditable scoring

1. Introduction

LLMs are being sold—implicitly or explicitly—as geopolitical analysts: rank theaters, fill ACH matrices, state confidence. Fluency makes that pitch easy. Tradecraft makes it false unless the model matches *inspectable* methods on the axes that decide whether a brief is usable.

This paper tests a dominance claim:

On fixed OSINT packs, ARGUS instruments dominate schema-locked LLMs on calibrated assessment tasks (absolute risk bands, ACH cells, confidence caps). LLMs match instruments only when the task reduces to following a rate rule or recovering rank order.

We do **not** claim instruments equal world ground truth. We claim something stronger for *systems*: instruments are **reproducible authorities**—same inputs, same numbers, no vibe. That is the standard LLMs fail here.

Contributions.

1. A locked, open replication package (research/llm-eval/).
 2. Head-to-head results: Claude Haiku 4.5 and GPT-4o-mini vs ARGUS instruments, four tasks × three packs.
 3. A **task-wise dominance map**: instruments win calibration; models win only the soft ordinal/numeric-follow tasks.
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2. Related work

Event data and risk scoring. Structured streams (ACLED, GDELT) and composites underpin political-risk practice. ARGUS implements absolute country risk in `lib/riskScoring.ts` / `lib/projectRisk.ts`.

ACH. Heuer’s method remains the falsification backbone of analytic tradecraft. We use ARGUS offline ACH (`lib/offlineIntel.ts`) as an auditable matrix—not human gold, but a fixed rule baseline LLMs must beat to claim ACH competence.

LLM evaluation. Schema adherence and calibration literature already warn against trusting fluent confidence. We operationalize that warning on geopolitical instruments.

Evidence completeness. Thin corpora must clamp confidence. ARGUS evidence-balance (`lib/evidenceBalance.ts`) emits an explicit HIGH/MODERATE/LOW cap.

3. Instruments (the authority set)

Task	Instrument	Authority output
Risk ranking	<code>projectRisk</code>	Score 0–100 + CRITICAL/HIGH/MEDIUM/LOW
Anomaly call	<code>rateZScore</code> + fixed thresholds	Surge + none/mild/strong
ACH matrix	<code>scoreACH0ffline</code>	supports / neutral / contradicts
Brief confidence	<code>assessEvidenceBalance</code>	<code>confidenceCap</code>

Thresholds were locked *ex ante*. Instruments are the reference; models are the challengers.

4. Experimental design

4.1 Datasets

Pack	Character	Events	Instrument anomaly
pack-conflict	Kinetic-heavy frontier	8	$z = 6.73$ strong
pack-maritime	Maritime + economic mix	4	$z = 1.73$ mild
pack-thin	Near-empty political corpus	2	$z = 0$ none

4.2 Challengers

Arm	Model	Effort
Claude	claude-haiku-4-5-20251001	low
OpenAI	gpt-4o-mini	low

Temperature 0; shared prompts/schemas; ≤ 12 events; token caps 800 (2000 for ACH).

4.3 Metrics (how dominance is scored)

Task	Model “wins” if...	Instrument dominates if...
Risk bands	High band agreement	Band agreement near zero / systematic inflation
Risk rank	High Spearman ρ	(secondary; ordinal only)
Anomaly	Surge + severity match	Model fails the rate rule
ACH	High cell agreement	Cell agreement far below usable matrix quality
Confidence	Cap match; no overclaim	Overclaim or miss the cap

5. Results

5.1 Instrument baselines (deterministic, \$0 API)

Pack	Top risk	Evidence cap	Anomaly
conflict-rich	Ukraine 46 / HIGH	MODERATE / 52	strong
maritime-mixed	Ukraine 14 / LOW	MODERATE / 63	mild
thin	Unknownland 1 / LOW	MODERATE / 63	none

5.2 Dominance table (primary claim)

Axis	Claude	GPT	Winner
Absolute risk bands	0.11 agree	0.11 agree	Instrument
Risk rank ρ	0.75	1.00	Model (ordinal only)
Anomaly surge / severity	1.00 / 1.00	1.00 / 1.00	Tie (models follow rate rule)
ACH cell agreement	0.40	0.53	Instrument
Confidence cap match	0.67	0.33	Instrument (Claude closer; GPT fails)
Overconfidence rate	0.00	0.33	Claude restrained; GPT loses

Reading the table. Models do not overturn the instruments on any *calibrated* axis. They either (a) fail bands and ACH, or (b) match anomalies by parroting a numeric comparison anyone could code. Rank correlation without band agreement is **false competence**: same order, wrong magnitude.

5.3 What failure looks like

GPT assigned Ukraine **CRITICAL** / **85** on the maritime pack while the instrument scored **LOW** / **14**. Same events; different authority. That is not a quibble—it is why LLM-only risk labels are unsafe.

ACH cell agreement of 0.40–0.53 means roughly half the falsification matrix disagrees with the offline method. That is not “almost ACH.”

GPT stated **HIGH** confidence on the conflict-rich pack against a **MODERATE** evidence cap—classic overclaim under fluency.

6. Discussion: where instruments dominate

1. **Bands and ACH define the product.** Briefs live or die on absolute levels and hypothesis cells. Instruments own both in this evaluation.
2. **Anomaly “wins” for models are cheap.** Matching z -thresholds shows schema following, not analytic superiority.
3. **Rank $\neq$ risk.** High Spearman with ~11% band agreement is the strongest *warning* result in the paper.
4. **Confidence is the tell.** Even when Claude avoids overclaim, it still does not justify removing the instrument cap.

Systems implication: ARGUS-style stacks should compute scores first and let models narrate second—not the reverse.

7. Limitations

- Pilot scale ($n = 3$ packs); larger corpora can only strengthen or weaken the map—they do not erase the band/ACH gap observed here.
 - Cost-tier models (Haiku / 4o-mini); if larger models close the gap, that is an empirical update, not a reason to skip instruments today.
 - Offline ACH $\neq$ human gold.
 - Synthetic, English, short-form events.
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8. Conclusion

Auditable instruments dominate language models on calibrated geopolitical assessment under this protocol. Claude and GPT match rate-based anomalies and can recover risk order; they do **not** match absolute bands or ACH cells at usable levels, and GPT overclaims confidence when the corpus looks rich. The correct architecture is instrument-authoritative scoring with optional LLM assistance—not LLM-authoritative scoring with optional formulas. Replication package: `research/llm-eval/`.

Replication

```
npm run eval:baselines
npm run eval:llm -- --provider openai
npm run eval:llm -- --provider claude
npm run eval:score
```

Artifacts: `results/tables/{baselines,scored,summary}.json, results/raw/*`.
Protocol: `PROTOCOL.md` · Decisions: `DECISIONS.md`.

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